76. With reference to coal-based thermal power plants in India, consider the following statements :

1. None of them uses seawater.

2. None of them is set up in water stressed district.

3. None of them is privately owned.

How many of the above statements are correct?

(a) Only one (b) Only two (c) All three (d) None

Ans. (d) : The Mundra Thermal Power Plant utilizes seawater drawn from Gulf Kutch. Thus statement (1) is incorrect.

According to recent study of World Resources Institute 40 percent India's thermal power plants are situated in regions experiencing significant water stress. Thus statement (2) is incorrect.

India has a total 269 thermal power plants of which 138 owned by public sector and remaining 131 by private. Thus statement (3) is incorrect.

77. 'Wolbachia method' is sometimes talked about with reference to which one of the following?

(a) Controlling the viral diseases spread by mosquitoes

(b) Converting crop residues into packing material

(c) Producing biodegradable plastics

(d) Producing biochar from thermo-chemical conversion of biomass

Ans. (a) : The 'Wolbachia method' is a technique used to control the spread of viral disease transmitted by mosquitoes, such as Dengue, Zika and Chikangunya. Wolbachia is a type of bacterium that can be introduced into mosquitoes to reduce their ability to transmit these diseases. When mosquitoes carrying Wolbachia mate with mosquitoes without Wolbachia the resulting eggs do not hatch or produce oppspring, thus reducing the population of disease carrying mosquitoes.

78. Consider the following activities :

1. Spreading finely ground basalt pick on farmlands extensively.

2. Increasing the alkalinity of oceans by adding lime.

3. Capturing carbon dioxide released by various industries and pumping it into abandoned subterranean mines in the form of carbonated waters

How many of the above activities are often considered and discussed for carbon capture and sequestration?

(a) Only one (b) Only two (c) All three (d) None

Ans. (c) : Spreading finely ground basalt rock on Farmlands accelerates natural process of carbon dioxide capture from the atmosphere. The basalt reacts with carbon dioxide to form stable carbonates, effectively storing carbon.

Increasing alkalinity of the ocean by addition of lime enhances the capacity of oceans to absorb and store carbon dioxide from the atmosphere thereby aiding in carbon sequestration.

Capturing carbon dioxide released by industries and pumping it into abandoned subterranean mines is carbon capture and storage method. Abandoned subterranean sites, with the captured carbon dioxide pumped into these locations in the form of carbonated waters.

79. 'Aerial metagenomics' best refers to which one of the following situations?

(a) Collecting DNA samples from air in a habitat at one go

(b) Understanding the genetic makeup of avian species of a habitat

(c) Using air-home devices to collect blood samples from moving animals

(d) Sending drones to inaccessible areas o collect plant and animal samples from land surfaces and water bodies

Ans. (a : 'Aerial met genomics' refers to the collection and analysis of DNA samples from the air or aerosol particles present in a specific habitat. It involves sampling the air in a particular environment to capture the genetic material (PDNA) in micro-organisms, including bacteria, viruses, Fungi and other microscopic organisms.

80. 'Microsatellite DNA' is used in the case of which one of the following?

(a) Studying the evolutionary relationships among various species of fauna

(b) Stimulating 'stem cells' to transform into diverse functional tissues

(c) Promoting clonal propagation of horticultural plants

(d) Assessing the efficacy of drugs by conducting series of drug trials in a population

Ans. (a) : Microsatellite DNA, also known as short tandem repeats, is a type of genetic marker that consist of repeating sequences of DNA. It is commonly used in genetic studies, including population genetics and phylogentic analysis, to study relationship among different species of Fauna.

81. Consider the following statements in relation to Janani Suraksha Yojana :

1. It is a safe motherhood intervention of the State Health Departments.

2. Its objective is to reduce maternal and neonatal mortality among poor pregnant women.

3. It aims to promote institutional delivery among poor pregnant women.

4. Its objective includes providing public health facilities to sick infants up to one year of age.

How many of the statements given above are correct.

(a) Only one (b) Only two (c) Only three (d) All four

Ans. (c) : Janani Suraksha Yojna is a safe motherhood intervention under the National Health Mission. It is not the intervention of State Health Departments.

The main objectives JSY are to reduce maternal and neonatal mortality by promoting institutional delivery among pregnant women especially with weak socio-economic status.

In 2014 the programme was extended to all antenatal & post-natal complications of all sick newborns and infants (upto one year of age) accessing public health institutions for treatment.

82. Consider the following statements in the context of interventions being undertaken under Anaemia Mukt Bharat Strategy :

1. It provides prophylactic calcium supplementation for pre-school children, adolescents and pregnant women.

2. It runs a campaign for delayed cord clamping at the time of childbirth.

3. It provides for periodic deworming to children and adolescents.

4. It addresses non-nutritional causes of anaemia in endemic pockets with special focus on malaria, hemoglobinopathies and fluorosis.

How many of the statements given above are correct?

(a) Only one (b) Only two (c) Only three (d) All four

Ans. (c) : Anaemia Mukt Bharat strategy provides prophylactic Iron folic acid supplement (not prophylactic calcium) for pre-school children, adolescents and pregnant women. It also runs a campaign for delayed cord clamping at the time of child birth. It provides for periodic dewarming to children and adolescents. It also addresses non nutritional causes of anemia endemic pockets with special focus on malaria hemoglobin pathies and flurosis.

83. Consider the following statements :

1. Carbon fibres are used in the manufacture of components used in automobiles and aircrafts.

2. Carbon fibres once used cannot be correct.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : Carbon Fibre is increasingly celebrated as a wonder material for the clean economy. Its unique combination of high strength and low weight has helped drive the wind power revolution and make planes more fuel efficient. Auto makers are also waking up to the materials potential to make lighter and more efficient vehicles.

Though difficult to recycle, carbon Fiber can be recycled.

Thus statement 1 is correct and statement 2 is incorrect.

84. Consider the following actions :

1. Detection of car crash/collision which results in the deployment of airbags almost instantaneously

2. Detection of accidental free fall of a laptop towards the ground which results in the immediate turning off of the hard drive

3. Detection of the tilt of the smart-phone which results in the rotation of display between portrait and landscape mode

In how many of the above actions is the function of accelerometer required?

(a) Only one (b) Only two (c) All three (d) None

Ans. (c) : Accelerometer are devices that measure the vibration or acceleration of motion of a structure.

Accelerometer is used in section of car collision/crash which results in the deployment of airbags almost instantaneously. If laptop were to suddenly drop while in use, the accelerometer would detect sudden free fall and immediately turn off the hard drive.

An accelerometer detect tilt of the Smartphone which results in rotation of display between portrait and landscape mode

Thus all three statements are correct.

85. With reference to the role of biofilters in Recirculating Aquaculture System, consider the following statements :

1. Biofilters provide waste treatment by removing uneaten fish food.

2. Biofilters convert ammonia present in fish waste to nitrate.

3. Biofilters increase phosphorus as nutrient for fish in water.

How many of the statements given above are correct?

(a) Only one (b) Only two (c) All three (d) None

Ans. (b) : Biofilters are important components of Recirculating Aquaculture systems to remove nitrogenous wastes excretes by aquatic organisms. Here nitrifying bacteria are used as biofilters to oxidise ammonia and convert it into nitrites which are also toxic. Some bacteria also convert these nitrites into final products which are non-toxic nitrates.

Biofilters are designed to filter pollutants and not to add nutrients. Thus statements 1 & 2 are correct and 3 is incorrect.

86. Consider the following pairs :

Objects in space			Description
1.	Cepheids	:	Grant clouds of dust and gas in space
2.	Nebulae	:	Stars which brighten and dim periodically
3.	Pulsars	:	Neutron stars that are formed when massive stars run out of fuel and collapse

How many of the above pairs are correctly matched?

(a) Only one (b) Only two

(c) All three (d) None

Ans. (a) : Cepheids are stars which brighten and dim periodically. Thus Pair-1 is incorrect.

A nebula is giant cloud of dust and gas in space. Thus pair 2 is incorrect.

Pulsars are neutron stars. Thus pair 3 is correct.

87. Which one of the following countries has its own Satellite Navigation System?

(a) Australia (b) Canada

(c) Israel (d) Japan

Ans. (d) : Quasi-Zenith satellite (8255) system is a regional satellite navigation system owned by Government of Japan.

88. Consider the following statements :

1. Ballistic missiles are jet-propelled at subsonic speeds throughout their flights, while cruise missiles are rocket-powered only in the initial phase of flight.

2. Agni-V is a medium-range supersonic cruise missile, while BrahMos is a solid-fuelled intercontinental ballistic missile.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : Cruise Missiles are jet propelled at subsonic speeds throughout their flights, while ballistic missiles are rocket-powered only in the initial phase of flight after which they follow an arching trajectory to the target.

Agni-V is a ballistic missile while the BRAHMOS is a medium range ramjet Supersonic cruise missile.

Thus both the statements are incorrect.

89. Consider the following statements regarding mercury pollution :

1. Gold mining activity is a source of mercury, pollution in the world.

2. Coal-based thermal power plants cause mercury pollution.